

Erdős Problem #875

EP-875

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A = \{a_1 < a_2 < \dots\} \subset \mathbb{N}$ be an infinite set such that the sets $S_r = \{a_1 + \dots + a_r : a_1 < \dots < a_r \in A\}$ are disjoint for distinct $r \geq 1$. How fast can such a sequence grow? How small can $a_{n+1} - a_n$ be? In particular, for which c is it possible that $a_{n+1} - a_n \leq n^c$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>